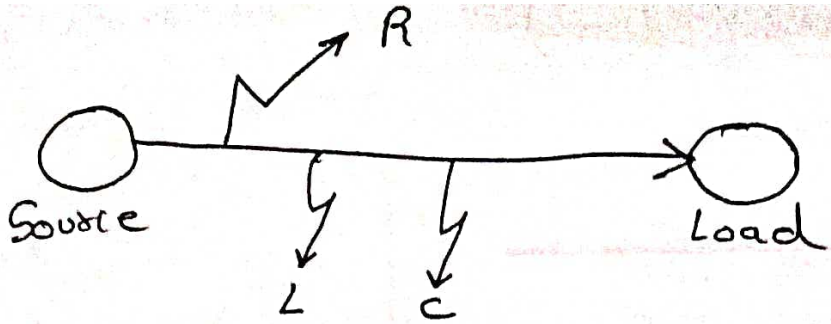
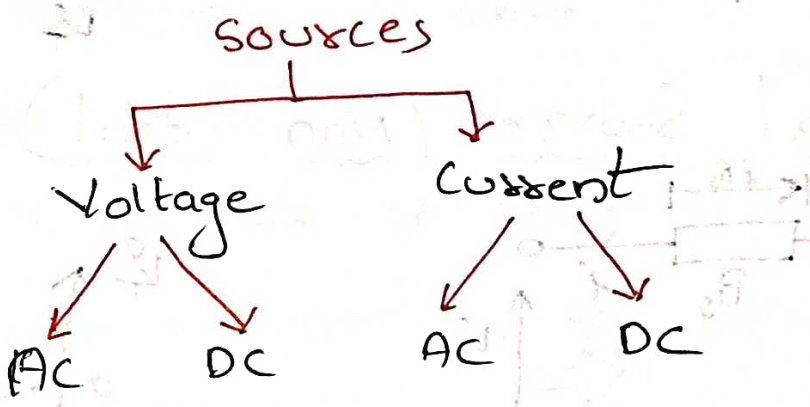
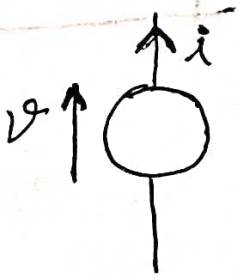


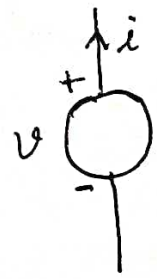
Electrical Sources:

1. Wall outlet
2. Batteries
3. Solar PV cell
4. Wind, hydro .....

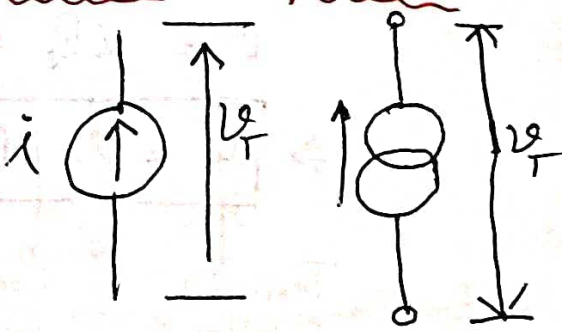


*. Voltage Source:

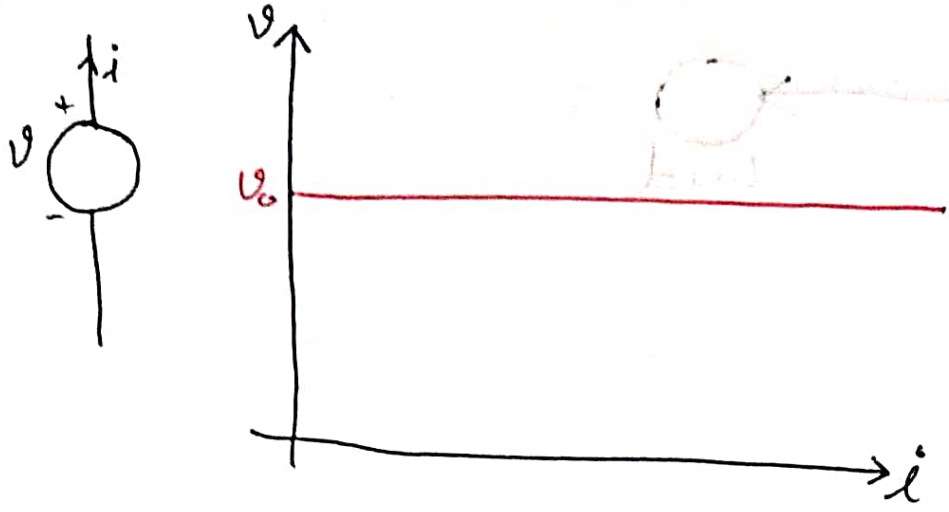


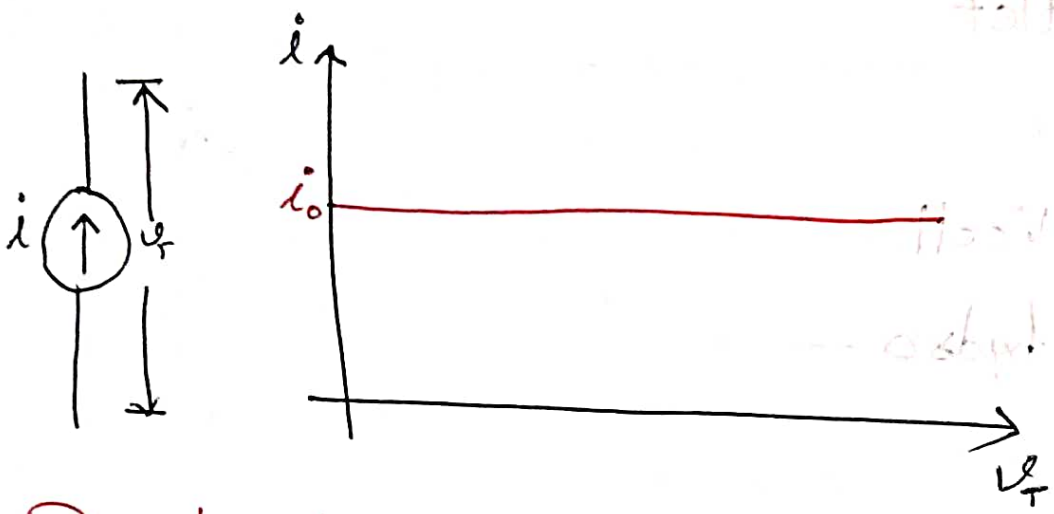


Current Source



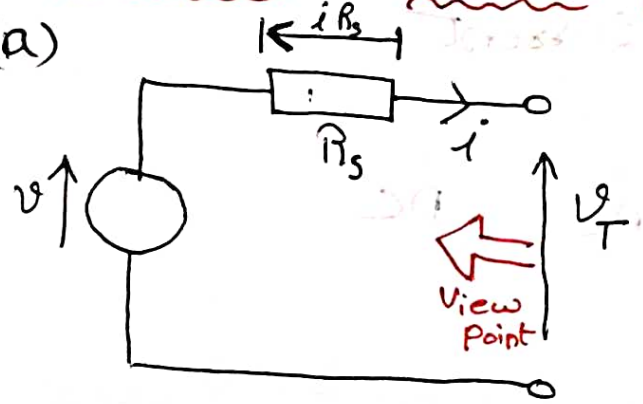
*. V- I characteristics (Ideal)

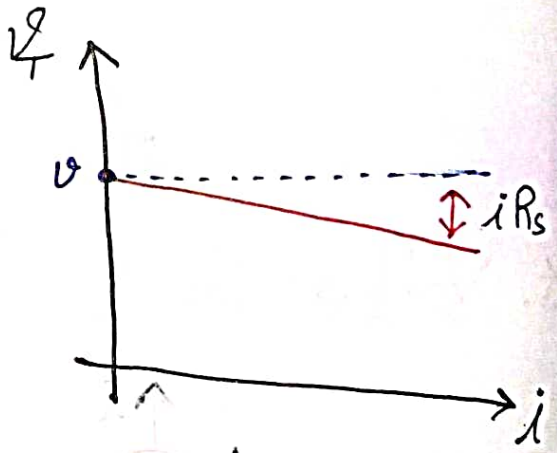




*. Practical Sources (Non-ideal):

(a)



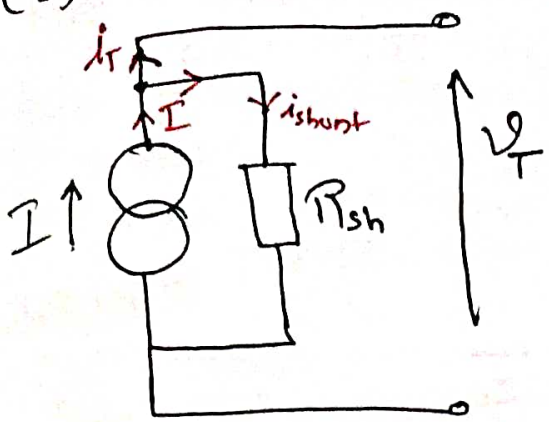


$$v = iR_s + v_t$$

$$\Rightarrow \boxed{v_t = v - iR_s}$$

Terminal Voltage — Load Current

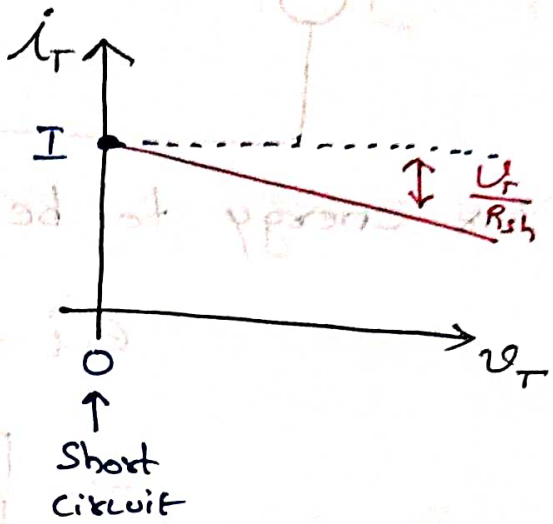
(b)



$$I = i_T + i_{shunt}$$

$$= i_T + \frac{v_T}{R_{shunt}}$$

$$\Rightarrow \boxed{i_T = I - \frac{v_T}{R_{sh}}}$$



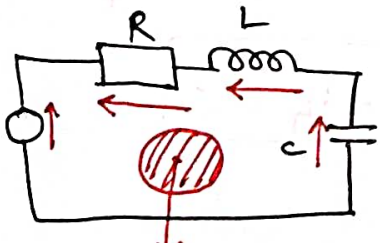
*. Kirchoff's Laws:

→ It operates on principle of conservation of energy.

→ Electronic circuits are made up of junctions.

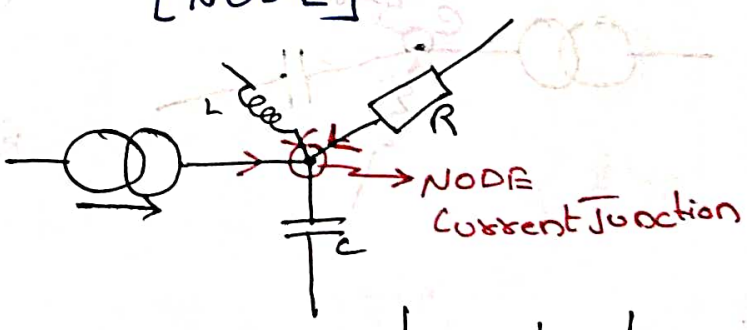
Junctions

- Voltage Junctions [LOOP]
- Current Junctions [NODE]



Loop Voltage Junction

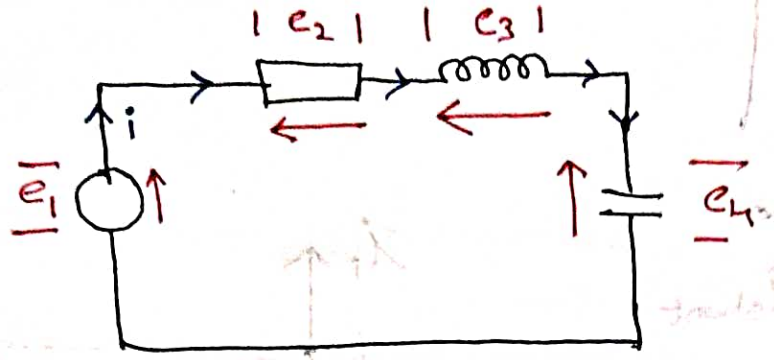
→ Current through loop is same



→ All the elements share the same Voltage

→ Both voltage junction & current junction are "ENERGY CONSERVING"

LOOP:-



For energy to be conserved

$$e_1 i = e_2 i + e_3 i + e_4 i$$

$$\Rightarrow \boxed{e_1 - e_2 - e_3 - e_4 = 0}$$

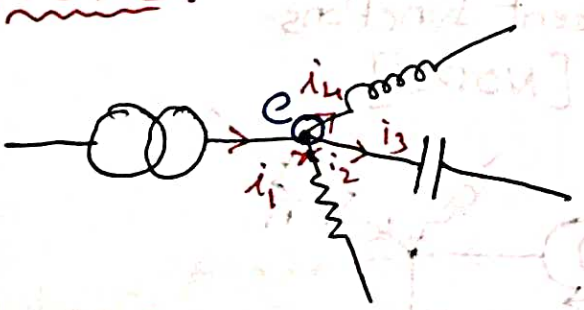
$$\Downarrow$$

$$\sum e_x = 0$$

∴ The sum of all voltages in a loop is zero.

This law is called "Kirchoff's Voltage Law (KVL)"

NODE:



$$e i_1 = e i_2 + e i_3 + e i_4$$

$$\Rightarrow \boxed{i_1 - i_2 - i_3 - i_4 = 0}$$

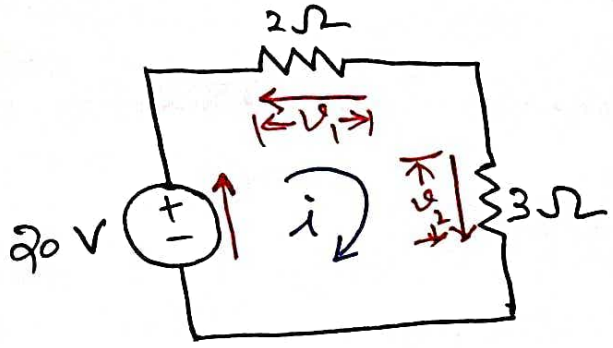
$$\Downarrow$$

$$\sum i_x = 0$$

"The algebraic sum of all currents at a node is zero. This law is called "Kirchoffs Current Law" (KCL)

Example 01:

For the circuit shown below, find voltages v_1 & v_2.



Sol:- To find v_1 & v_2 apply ohms law & KVL

$v_1 = 2i$; $v_2 = -3i$; ⇨ Ohms law

- Applying KVL around the loop gives

$$20 - v_1 + v_2 = 0$$

$$\Rightarrow 20 = v_1 - v_2$$

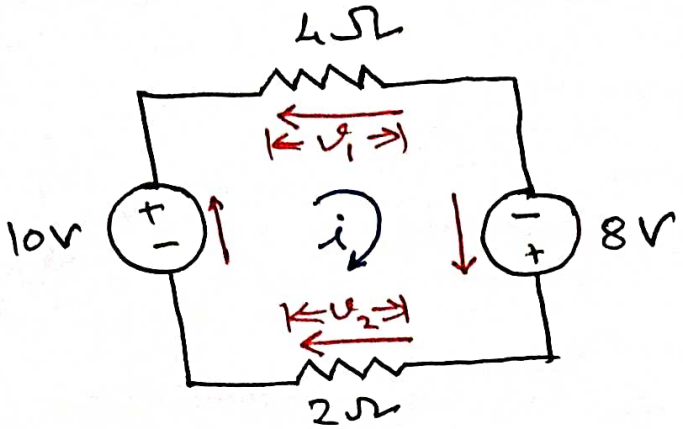
$$\Rightarrow 20 = 5i \Rightarrow \boxed{i = 4A}$$

∴ substitute the value of 'i' in 'v_1' & 'v_2'

$$\therefore v_1 = 8\,V; \quad v_2 = -12V;$$

Example: 02

Find v_1 & v_2 for the circuit shown below



Sol:- Apply ohms law for resistors

$$v_1 = 4i \; ; \; \& \; v_2 = 2i$$

Applying KVL around loop yields

$$\Rightarrow 10 - 4i + 8 - 2i = 0$$

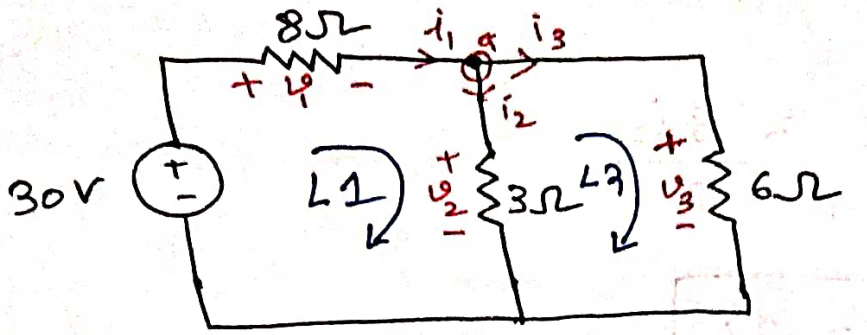
$$\Rightarrow 18 = 6i \Rightarrow \boxed{i = 3A}$$

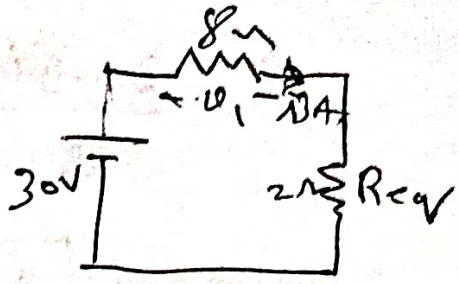
Substitute the value of 'i' to yield 'v_1' & 'v_2'

$$\therefore v_1 = 12V \; ; \; v_2 = 6V$$

Example: 03

Find the currents and voltages in the circuit as shown below.





$\frac{3\times6}{9} = 2\Omega$

Sol:- Apply Ohms law:

$v_1 = 8i_1$; $v_2 = 3i_2$; $v_3 = 6i_3$; 30 − 24 − 6 =

At node 'a': Apply KCL

$$i_1 = i_2 + i_3 \longrightarrow (1)$$

Applying KVL to loop 1 (L1)

$$30 - 8i_1 - 3i_2 = 0$$

$$\Rightarrow i_1 = \frac{30 - 3i_2}{8} \longrightarrow (2)$$

Applying KVL to Loop 2 (L2)

$$-v_2 + v_3 = 0$$

$$\Rightarrow v_2 = v_3 \longrightarrow (3)$$

$$\Rightarrow 3i_2 = 6i_3$$

$$\Rightarrow \boxed{i_2 = 2i_3} \text{ (or) } \boxed{i_3 = \frac{i_2}{2}} \longrightarrow (3)$$

By substituting (3) & (2) in (1)

$$\Rightarrow \quad \frac{30-3i_2}{8} = i_2 + \frac{i_2}{2}$$

$$\Rightarrow \quad \frac{30-3i_2}{8} = \frac{3i_2}{2}$$

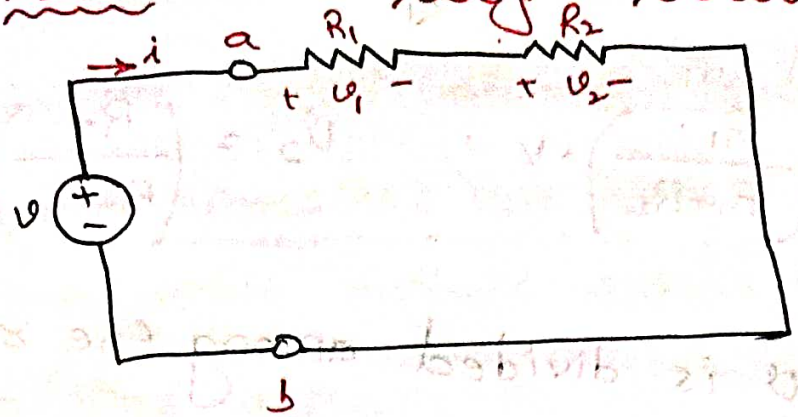
$$\Rightarrow \quad \boxed{i_2 = 2A}$$

⇒ Now substitute i_2 to get i_1 & i_2

$i_1 = 3A$, $i_2 = 2A$, $i_3 = 1A$,

$$\Rightarrow \quad \boxed{\begin{aligned} v_1 &= 24\ V \\ v_2 &= 6\ V \\ v_3 &= 6\ V \end{aligned}}$$

Series Resistors and Voltage division:



⇒ In the above circuit, R_1 & R_2 are connected in series.

⇒ The current flowing in both resistors are same

$$\therefore v_1 = iR_1 \; ; \; v_2 = iR_2$$

By applying KVL,

$$\Rightarrow v - v_1 - v_2 = 0$$

$$\Rightarrow v - iR_1 - iR_2 = 0$$

$$\Rightarrow v = i(R_1 + R_2)$$

$$\Rightarrow \boxed{i = \frac{v}{(R_1 + R_2)}}$$

Let

$$R_1 + R_2 = R_{eq}$$

$$\therefore \boxed{v = i \cdot R_{eq}}$$

For N resistors in series

$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^{N} R_n$$

Thus equivalent resistance of any number of resistors connected in series is the sum of the individual resistances.

→ To determine the voltage across each resistor

$$v_1 = \left(\frac{R_1}{R_1+R_2}\right)\cdot V \; ; \quad v_2 = \left(\frac{R_2}{R_1+R_2}\right)\cdot v$$

⇒ the voltage v is divided among the resistors in direct proportion to resistances.

→ i.e., larger the resistance → larger the drop

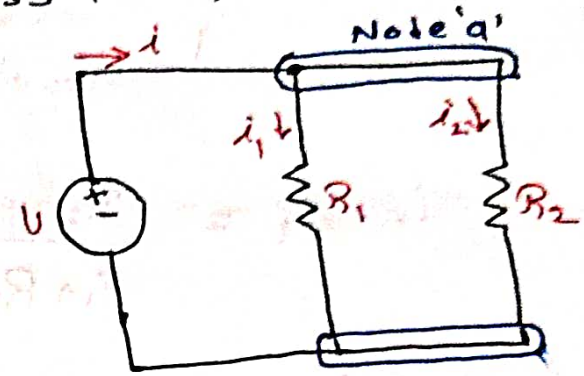
⇒ This is called "principle of voltage division" and the circuit is called "voltage divider"

⇒ In general, if a voltage divider has 'N' resistors $(R_1, R_2, \cdots R_N)$ in series, with the source voltage v, the nth resistor (R_n) will have a voltage drop of

$$v_n = \frac{R_n}{R_1+R_2+\cdots+R_N}\cdot v$$

Parallel Resistors and Current Division:

Consider the circuit shown in the following figure where two resistors are connected in parallel and therefore same voltage across them.



∴ From ohms law:

$$v = i_1 R_1 = i_2 R_2$$

$$\Rightarrow i_1 = \frac{v}{R_1} \; ; \quad i_2 = \frac{v}{R_2}$$

By applying KCL at node 'a'

$$i = i_1 + i_2$$

$$= \frac{v}{R_1} + \frac{v}{R_2}$$

$$= v\left[\frac{1}{R_1} + \frac{1}{R_2}\right] = \frac{v}{R_{eq}}$$

where R_{eq} is the equivalent resistance of the resistors in parallel.

$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\Rightarrow \boxed{R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}}$$

⇒ The equivalent resistance of two parallel resistors is equal to the product of their resistances

divided by their sum.

$$\therefore i_1 = \frac{R_2}{R_1 + R_2}$$

$$v = i R_{eq} = i \cdot \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore i_1 = \frac{R_2}{R_1 + R_2} \cdot i \quad ; \quad i_2 = \frac{R_1}{R_1 + R_2} \cdot i$$

⇒ The total current (i) shared by the resistors is inverse proportional to their resistances.

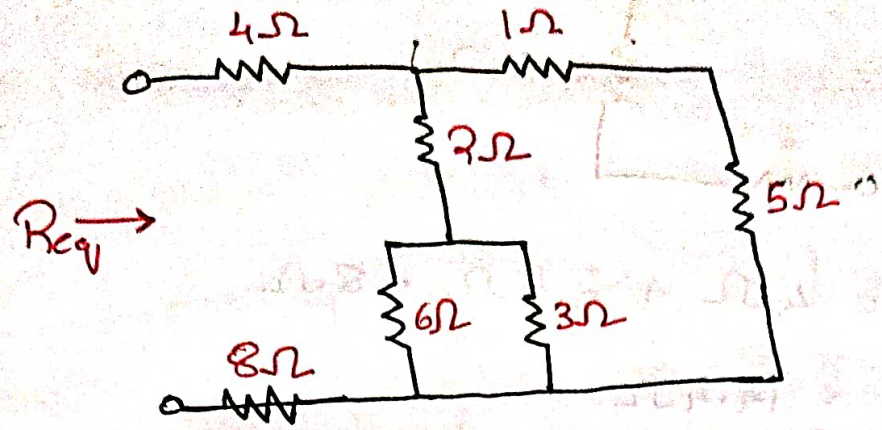
This is known as "principle of current division".

This circuit is known as "current divider".

Note: Larger current flows through smaller resistance.

Example 01:

Find R_{eq} for the circuit shown in below figure



Sol:- (1)

$6\Omega \,||\, 3\Omega$

$\therefore R_{6,3} = \frac{6\times 3}{9}$

$= 2\Omega$

(2)

1Ω — 5Ω

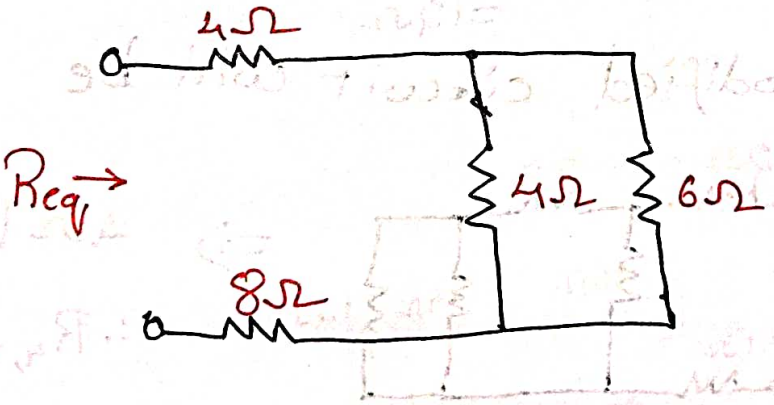
$R_{1,5} = 1+5$

$= 6\Omega$

(3) 2Ω — $(6\Omega \,||\, 3\Omega)$

$\Rightarrow R_{2,63} = 2\Omega + 2\Omega = 4\Omega$

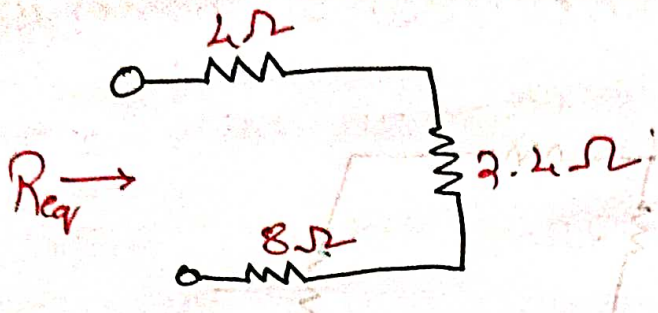
$\therefore$ The circuit will be modified as



$4\Omega \,||\, 6\Omega$

$\therefore R_{4,6} = \frac{4\times 6}{10} = 2.4\Omega$

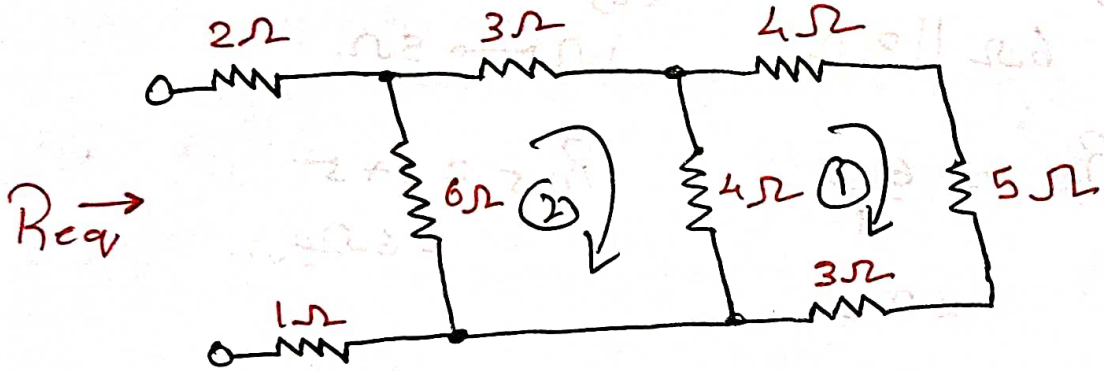
∴ The circuit further modifies



$\therefore R_{eq} = 4\Omega + 2.4\Omega + 8\Omega$

$= 14.4\Omega$

Example: 02



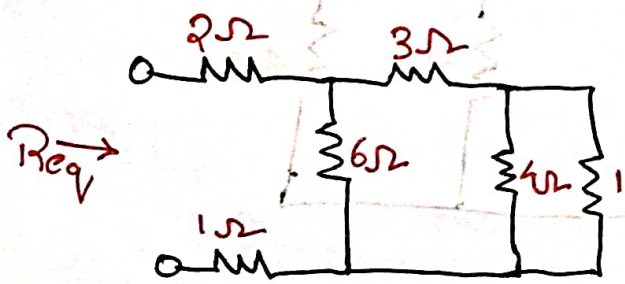
Sol:- Consider LOOP 1:



$\therefore R_{1eq} = 4 + 5 + 3$

$= 12\Omega$

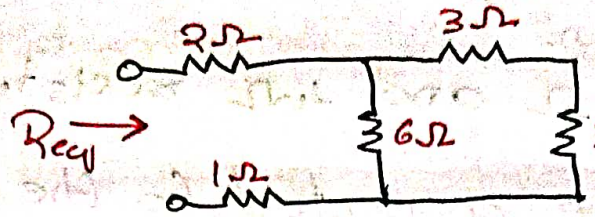
The modified circuit will be



$\Rightarrow 4\Omega \,||\, 12\Omega$

$\therefore R_{4,2} = \frac{4 \times 12}{16} = 3\Omega$

The modified circuit will be

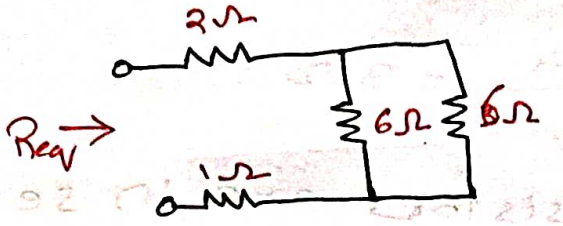


$\Rightarrow$ 3Ω o—o 3Ω

$\Rightarrow R_{2eq} = 3\Omega + 3\Omega$

$= 6\Omega$

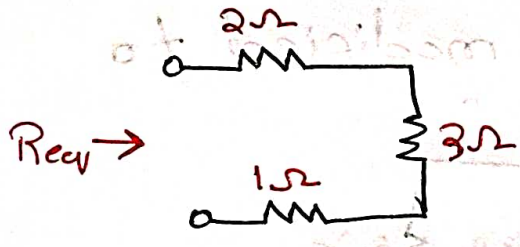
$\Rightarrow$ The modified circuit



$\Rightarrow 6\Omega \| 3\Omega$

$\Rightarrow R_{6,3} = \frac{6\Omega \times 6\Omega}{12\Omega} = 3\Omega$

$\Rightarrow$ The circuit is modified as

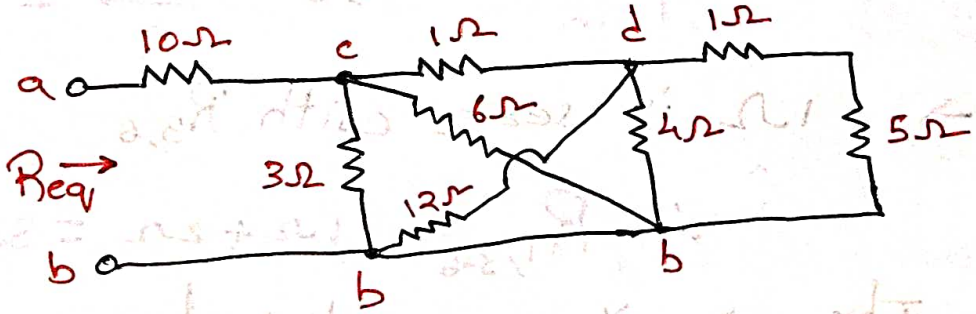


$\Rightarrow R_{eq} = 2\Omega + 3\Omega + 1\Omega$

$= 6\Omega$

Example: 03

Calculate the equivalent resistance R_{ab} in the circuit shown below.



Sol:- The 3Ω & 6Ω resistors are in parallel because they are connected to the same two nodes c and b.

$$\therefore R_{6,3} = \frac{6\Omega \times 3\Omega}{9\Omega^2} = 2\Omega$$

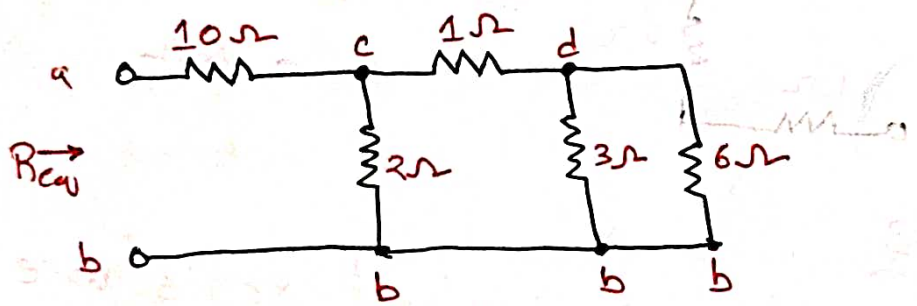
Similarly, the 12Ω and 4Ω resistors are in parallel since they are connected to the same nodes 'a' and 'b'.

$$\therefore R_{12,4} = \frac{12 \times 04}{16} = 3\Omega$$

Also 1Ω and 5Ω resistors are in series

$$\therefore R_{1,5} = 6\Omega$$

→ The circuit will be modified to



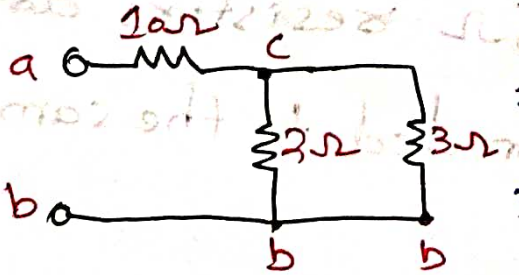
→ From the above circuit

$$3\Omega \| 6\Omega \Rightarrow R_{3,6} = \frac{3 \times 6}{9} = 2\Omega$$

→ 1Ω is series with $R_{3,6}$

$$\therefore R_{1,3\text{-}6} = 1\Omega + 2\Omega = 3\Omega$$

∴ The circuit is modified as,

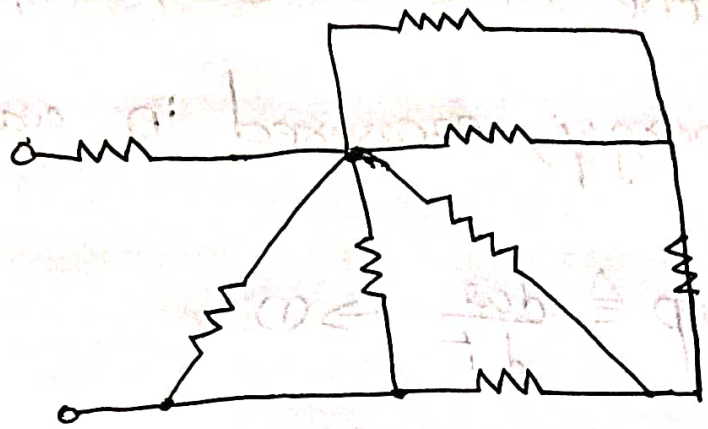


$$\Rightarrow (2\Omega \| 3\Omega) + 10\Omega = Req$$

$$\Rightarrow Req = 10 + \left[\frac{2 \times 3}{5}\right] = 11.2\Omega$$

Example 04:

Find R_{ab} for the circuit shown below



Power & Energy

→ Power is the time rate of expanding or absorbing energy, measured in watts (W).

$$p \triangleq \frac{dw}{dt} \rightarrow (1)$$

where

p = power in watts (W)

w = energy in joules (J)

t = time in seconds (s)

(1) can be re written as

$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v \cdot i$$

$$\therefore \boxed{p = v \cdot i} \rightarrow (2)$$

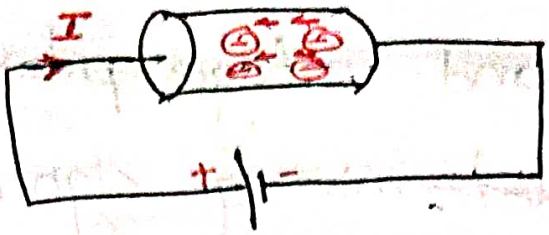
Note!!

Voltage:

To move electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by "Electro Motive Force (EMF)"

$$v_{ab} \triangleq \frac{dw}{dq}$$

Current:



→ When a conducting wire is connected to a battery, the charges are compelled to move in one direction. This motion of charges creates "electric current".

→ It is conventional to consider current flow as movement of positive charges.

$$i \triangleq \frac{dq}{dt}$$

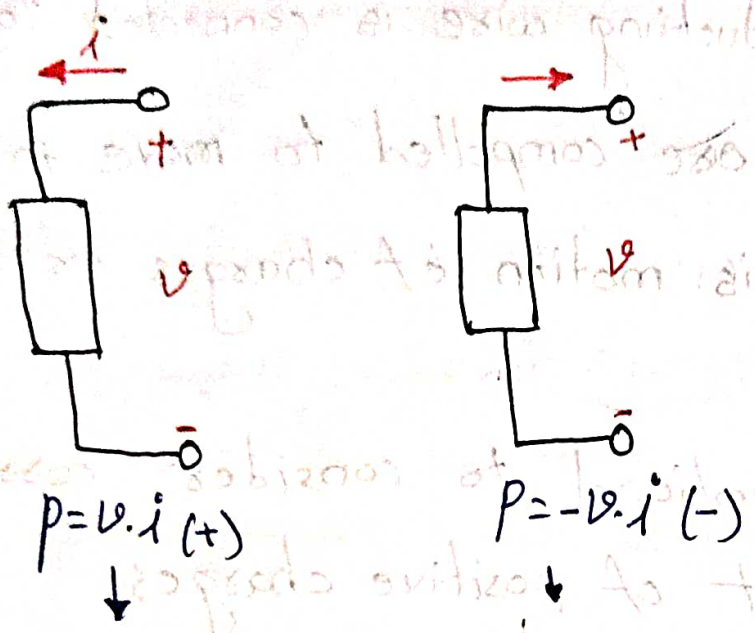
→ In (2) p is time varying quantity and is called "instantaneous power".

→ If P → + sign : power is being delivered to (or) absorbed by the element

→ If P → - ve sign : Power is being supplied by the element.

How to determine the sign of Power?

→ Current direction and voltage polarity are major factors for determining sign of power.



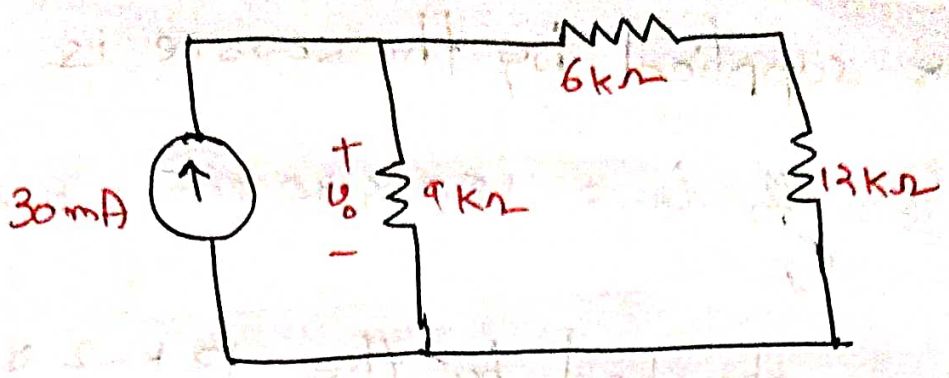
Example 01:

For the circuit shown below, determine

(a) voltage v_o

(b) power supplied by current source

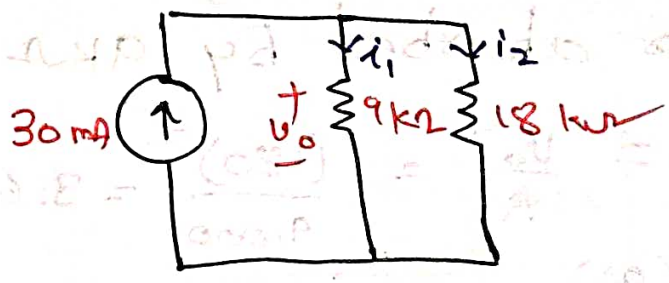
(c) power absorbed by each resistor.



Sol: 6kΩ & 12kΩ are in series and therefore

$$R_{6,12} = 6\text{k}\Omega + 12\text{k}\Omega$$
$$= 18\text{k}\Omega$$

Now the circuit modifies to



∴ By current division rule:

$$i_1 = \frac{18\times10^3}{9000+18000} \cdot 30\times10^{-3} = 20\text{mA}$$

$$i_2 = \frac{9{,}000}{9000+18000} = 10\text{mA}$$

∴ the voltage v_o is

$$v_o = 9000 \times i_1 = 180\text{V}$$

(b) Power supplied by the source is

$$P_0 = v_0 \cdot i_0 = 180 \times 30 \times 10^{-3} = 5.4\,W$$

(c) Power absorbed by the 12 K-Ω resistor is

$$P = iv = i_2^2 R = (10 \times 10^{-3})^2 (12{,}000)$$

$$= 1.2\,W$$

Power absorbed by the 6 kΩ resistor is

$$P = i_2^2 R = (10 \times 10^{-3})^2 \times (6000) = 0.6\,W$$

Power absorbed by 9kΩ resistor is

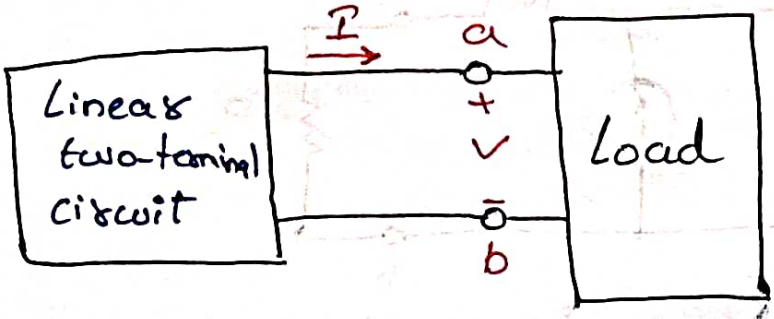
$$p = \frac{v_0^2}{R} = \frac{(180)^2}{9{,}000} = 3.6\,W$$

(or)

$$P = v_0 \cdot i_1 = 180\,(20)\,mW = 3.6\,W$$

Thevenin's Theorem:

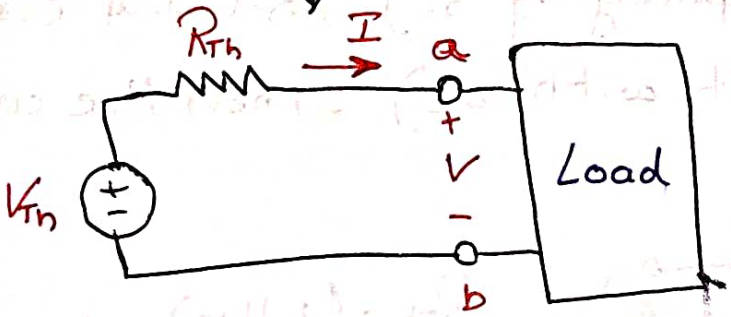
"A linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th}".



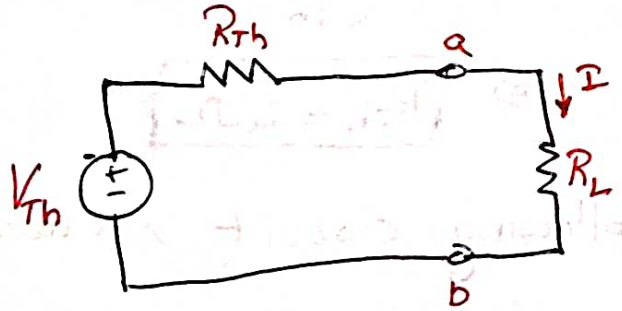
where

V_{Th} = Open circuit voltage at the terminals

R_{Th} = equivalent resistance at the terminals



⇒ ∴ The circuit modifies to

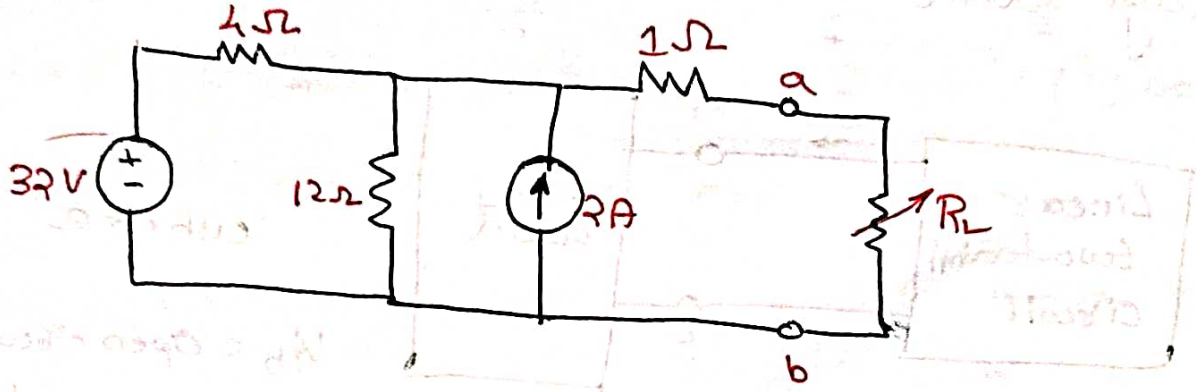


$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

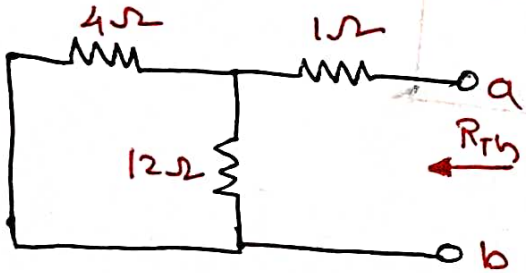
$$V_L = I_L R_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

Example 01

Find the Thevenin equivalent circuit of the circuit shown in below figure, to the left of terminals a-b. Then find the current through R_L = 6, 16 and 36Ω.



Step: 1 Calculate R_{Th} by turning OFF 32-V voltage source (replacing by sc) and 2-A current source (replacing it with oc). Then the circuit modifies to

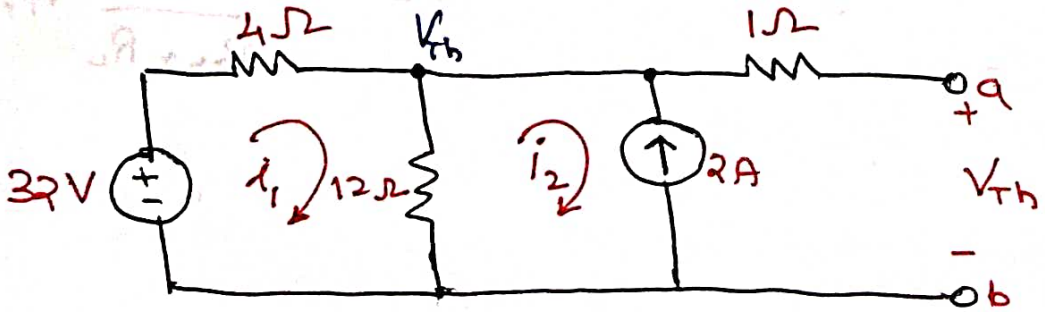


$$R_{Th} = (4 \| 12) + 1$$

$$= \left(\frac{4 \times 12}{16}\right) + 1$$

$$\Rightarrow \boxed{R_{Th} = 4\Omega}$$

Step: 2 Consider the following circuit for calculating V_{Th}



Apply mesh analysis to the two loops,

$$32 - 4i_1 - 12(i_1 - i_2) = 0 ; \quad i_2 = 2A$$

$$\Rightarrow 32 = 4i_1 + 12i_1 + 12i_2$$

$$\Rightarrow 32 - 12i_2 = 16i_1$$

$$\Rightarrow i_1 = \frac{32 - (12 \times 2)}{16} = \frac{8}{16} = 0.5A$$

$$\therefore V_{Th} = 12(i_1 - i_2)$$

$$= 12(0.5 + 2.0) = 30V$$

Nodal Analysis:

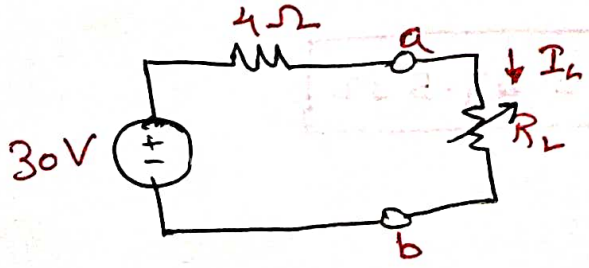
at Node V_{Th}

$$\frac{32 - V_{Th}}{4} + 2 = \frac{V_{Th}}{12}$$

$$\Rightarrow 96 - 3V_{Th} + 24 = V_{Th}$$

$$\Rightarrow \boxed{V_{Th} = 30V}$$

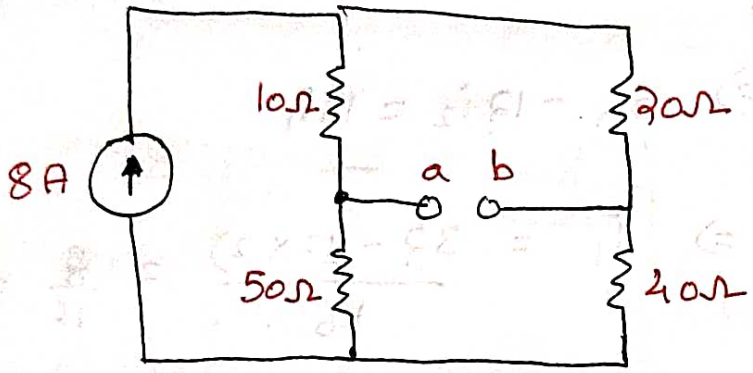
Step: 3 Find Thevinin equivalent circuit.



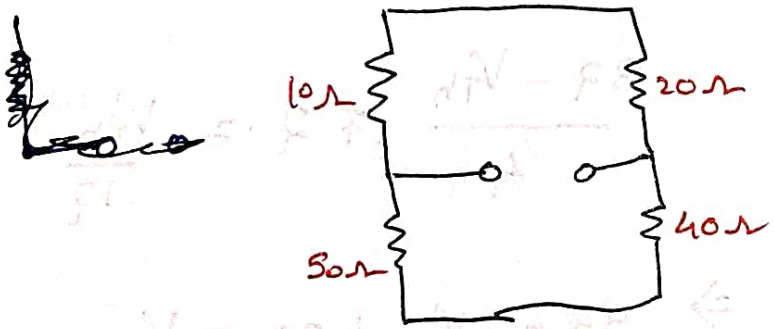
$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

Example : 02

Determine the Thevenin equivalent at terminals a-b for the circuit shown below:



Step: 1 Calculate the R_{Th} by turning OFF 8A current source (open circuit). Then the circuit modifies to

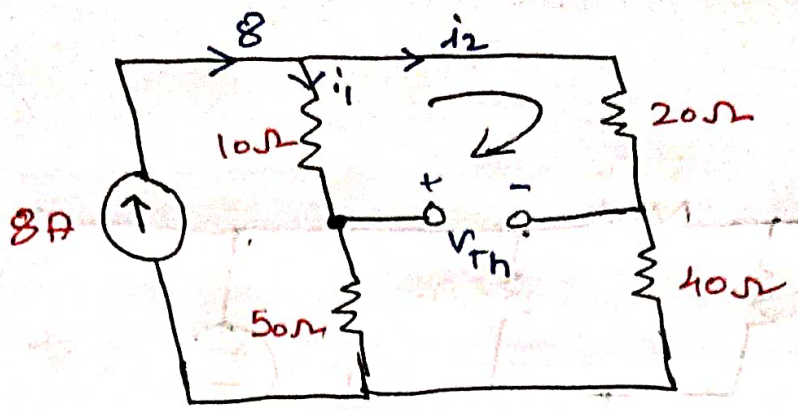


$$\therefore R_{Th} = (10+20) \,||\, (50+40)$$

$$= 30 \,||\, 90 = \frac{30 \times 90}{120}$$

$$\Rightarrow \boxed{R_{Th} = 22.5\ \Omega}$$

Step 2: Calculate the V_{Th}



$$i_1 = \frac{8 \cdot 60}{120} = \frac{8}{2} = 4\,A$$

$$i_2 = 8 \cdot \frac{60}{120} = \frac{8}{2} = 4A$$

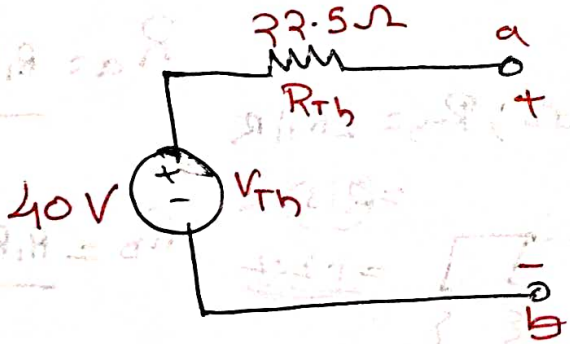
→ Now Apply KVL b/w a & b

$$-20 i_2 + V_{Th} + 10 i_1 = 0$$

$$\Rightarrow V_{Th} = 20 i_2 - 10 i_1 = 40\,V$$

$$\therefore \boxed{V_{Th} = 40\,V}$$

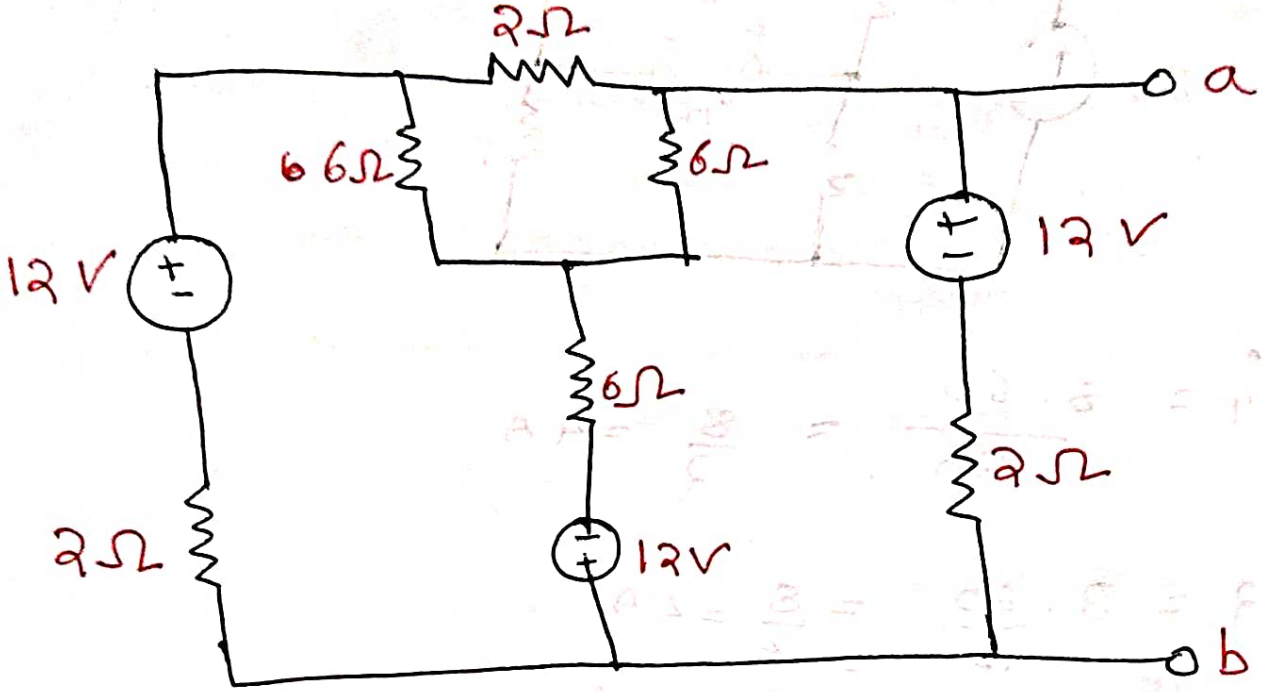
∴ The thevinin equivalent circuit is



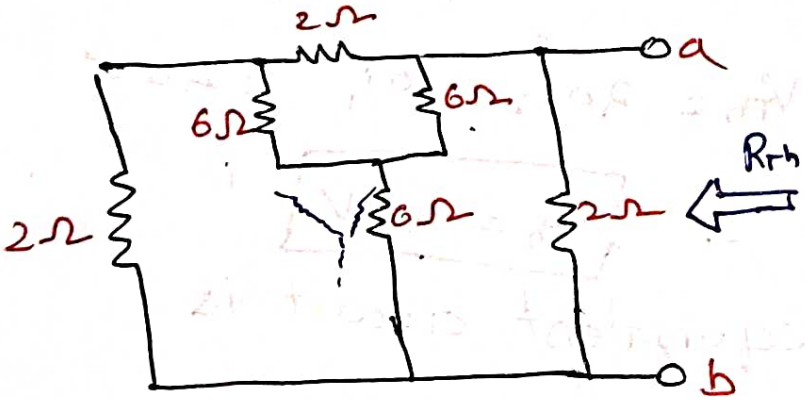
Example: 03:

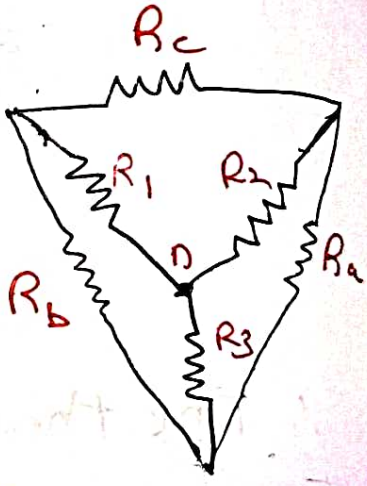
Obtain Thevinin equivalent circuit at termin

a-b:



Step 1: Calculate R_{Th}:

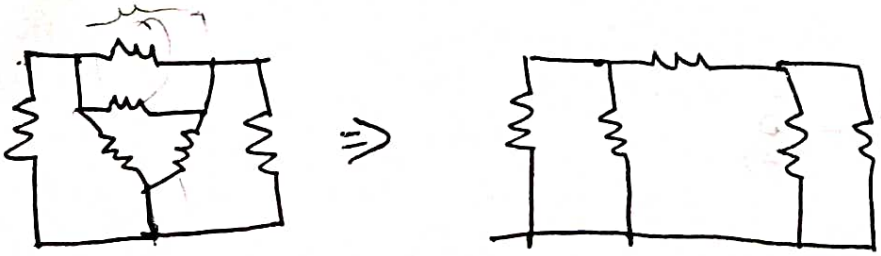




$R_{eq} = 2 \| 18 = 1.8\Omega$; $R_{Th} = 2R \| R$

$= (2 \| 3) R$

$= 1.2\Omega$

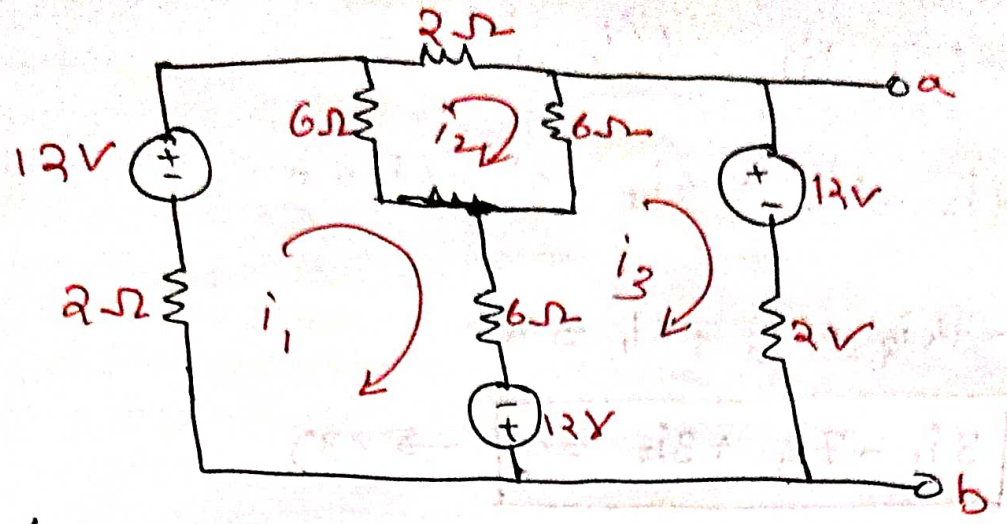


$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

Step: 2 Calculate V_{th}



Loop 1

$$12 - 6(i_1 - i_2) - 6(i_1 - i_3) - 2i_1 + 12 = 0$$

$$\Rightarrow 12 - 6i_1 + 6i_2 - 6i_1 + 6i_3 - 2i_1 + 12 = 0$$

$$\Rightarrow 24 + 6i_2 + 6i_3 = 14 i_1$$

$$\Rightarrow \boxed{7i_1 - 3i_2 - 3i_3 = 12} \rightarrow (1)$$

Loop 2:

$$-12 - 6(i_3 - i_1) - 6(i_3 - i_2) - 12 - 2i_3 = 0$$

$$\Rightarrow -12 - 6i_3 + 6i_1 - 6i_3 + 6i_2 - 12 - 2i_3 = 0$$

$$\Rightarrow -24 + 6i_1 + 6i_2 = 14 i_3$$

$$\Rightarrow \boxed{7i_3 - 3i_1 - 3i_2 = -12} \rightarrow (2)$$

Loop 3:

$$-2i_2 - 6(i_2 - i_3) - 6(i_2 - i_1) = 0$$

$$\Rightarrow -2i_2 - 6i_2 + 6i_3 - 6i_2 + 6i_1 = 0$$

$$\Rightarrow -14i_2 + 6i_3 + 6i_1 = 0$$

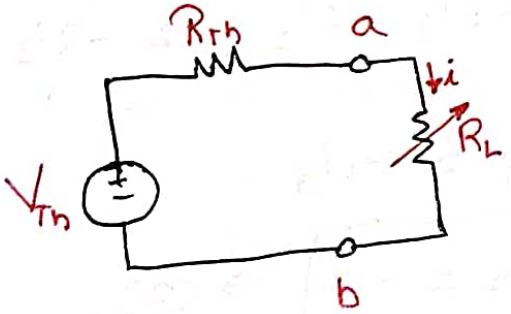
$$\Rightarrow \boxed{3i_1 - 7i_2 + 3i_3 = 0} \rightarrow (3)$$

Maximum Power Transfer:

→ These are applications in areas such as communications where it is desirable to maximize the power delivered to load.

→ It should be noted that this will result in significant internal losses greater than or equal to the power delivered to the load.

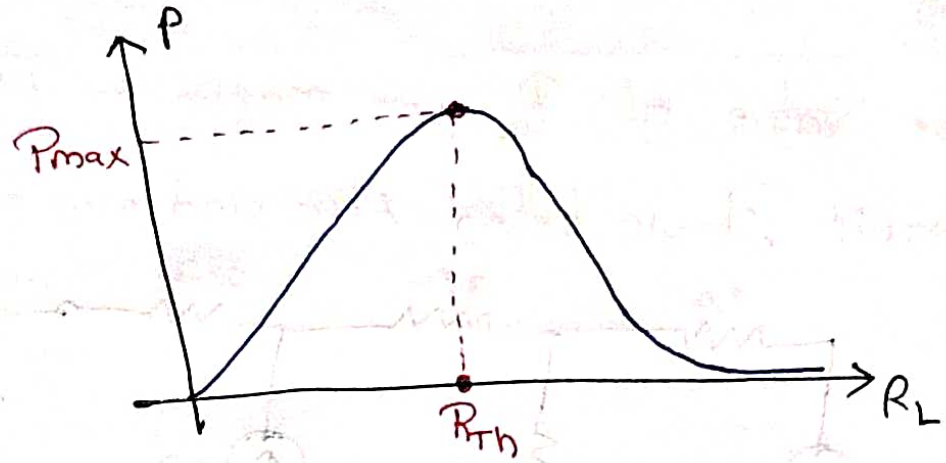
⇒ Consider the Thevenin equivalent circuit:



$$\therefore P = i^2 R_L = \left(\frac{V_{Th}}{R_L + R_{Th}}\right)^2 \cdot R_L$$

⇒ For a given circuit, V_{Th} and R_{Th} are fixed.

⇒ By varying the load resistance R_L, the power delivered to load varies as sketched in below figure.



⇒ to find maximum power differentiate P with respect to R_L and set equal to zero.

$$\frac{dP}{dR_L} = V_{Th}^2 \left[\frac{(R_L+R_{Th})^2 - 2R_L(R_L+R_{Th})}{(R_L+R_{Th})^4} \right]$$

$$= V_{Th}^2 \left[\frac{R_L+R_{Th}-2R_L}{(R_L+R_{Th})^3} \right] = 0$$

$$\Rightarrow 0 = R_L + R_{Th} - 2R_L = R_{Th} - R_L$$

which yields

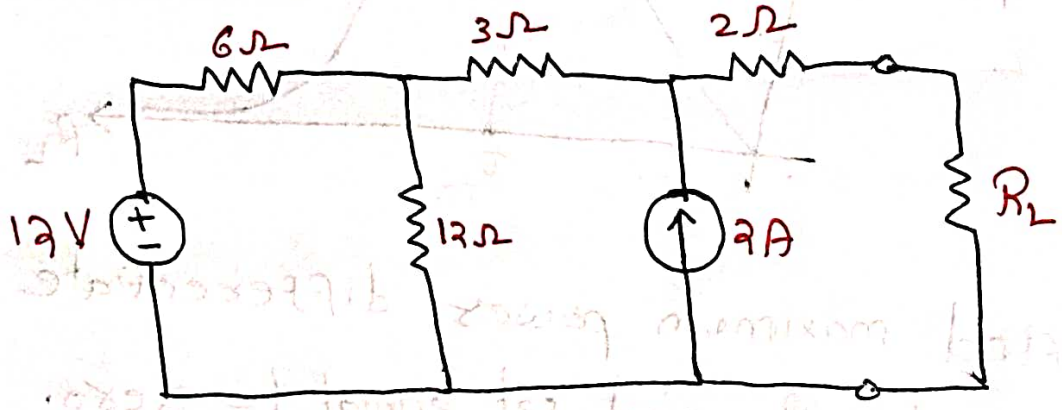
$$\boxed{R_L = R_{Th}}$$

∴ Maximum power is transferred to the load when the load resistance equals Thevenin's resistance as seen from load.

$$\therefore \quad P_{max} = \frac{V_{Th}^2}{4 R_{Th}}$$

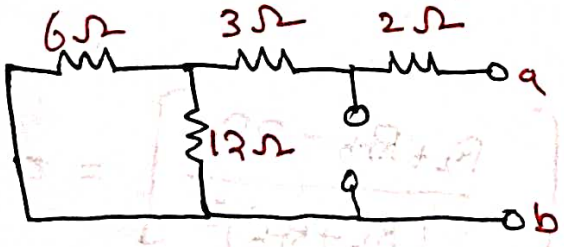
Example: 01

Find the value of R_L for maximum power transfer in the circuit shown below. Also find maximum power.



Step 1: Calculate V_{Th} & R_{Th}

(i) R_{Th}:

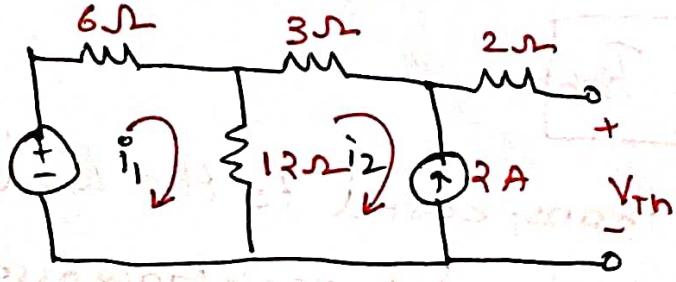


$$R_{Th} = (6 \| 12) + 3 + 2$$

$$= 5 + \frac{6 \times 12}{18}$$

$$\Rightarrow \boxed{R_{Th} = 9\Omega}$$

(ii) V_{Th}:



Loop-1:

$\Rightarrow 12 - 6i_1 - 12(i_1 - i_2) = 0$

$\Rightarrow 12 - 6i_1 - 12i_1 + 12i_2 = 0$

$\Rightarrow 12 - 18i_1 + 12i_2 = 0$

$\Rightarrow 18i_1 - 12i_2 = 12 \rightarrow (1)$ & $i_2 = -2\,A$;

$\therefore \boxed{i_1 = -\frac{2}{3}}$

Loop-2: (Outerloop)

$+12 - 3i_2 + V_{Th} = 0$
$+6i_1$

$\Rightarrow \boxed{V_{Th} = 22V}$

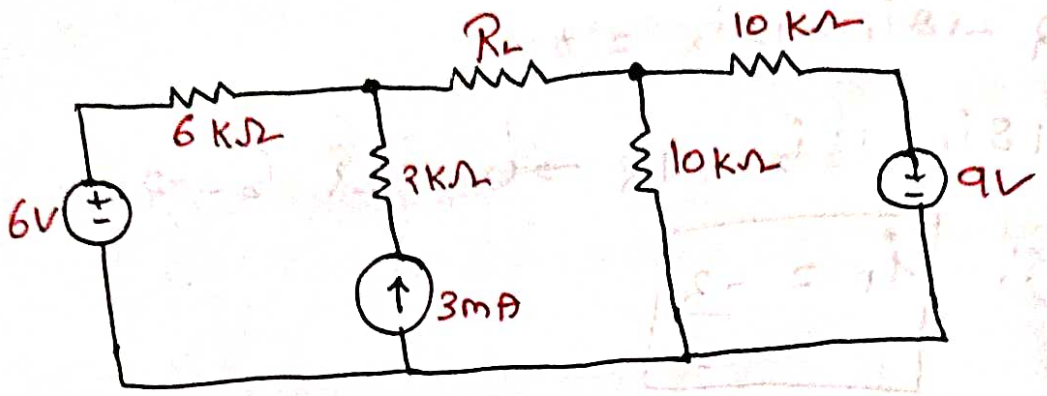
$\therefore$ For maximum power $R_L = R_{Th} = 9\Omega$

&

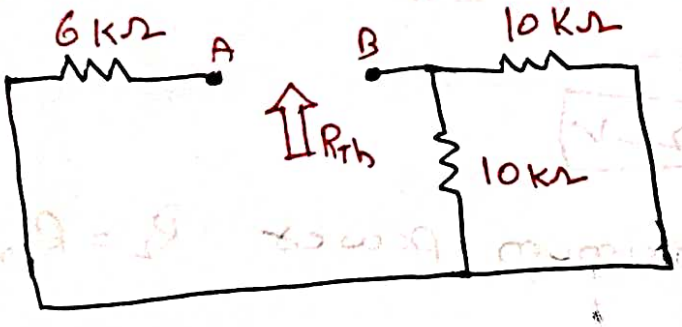
maximum power $= P_{max} = \frac{V_{Th}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44\,W$

Example 03:-

Calculate the maximum power that can be transferred to R_L in the below figure:



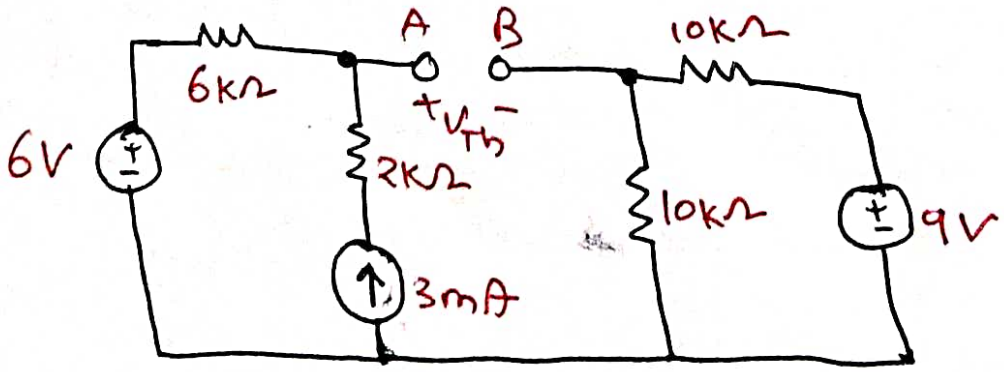
Step: 1 Calculate R_{Th}:



$$R_{Th} = 6\,K\Omega + \frac{(10\,K\Omega)(10\,K\Omega)}{20\,K\Omega}$$

$$\Rightarrow \boxed{R_{Th} = 11\,K\Omega}$$

Step:-2 Calculate V_{Th}:

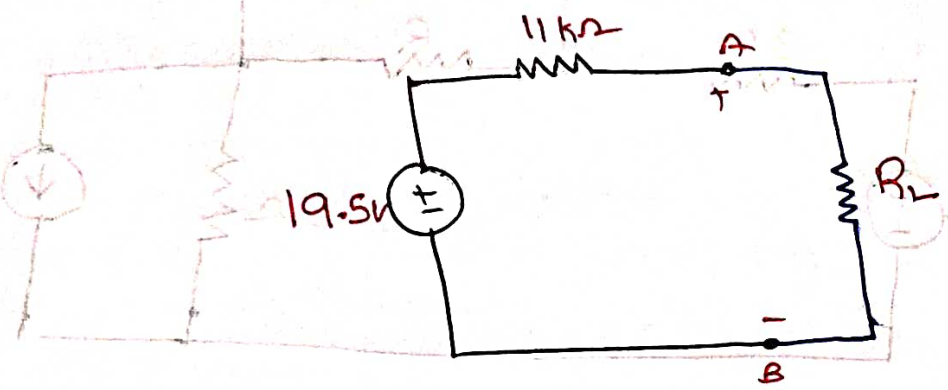


$\therefore$ V_{Th}:

$$6 + 18 - 4.5 - V_{Th} = 0$$

$$\Rightarrow \boxed{V_{Th} = 19.5\,V}$$

$\therefore$ Thevienin Equivalent ckt is



To gain maximum power:

$$R_L = R_{Th} = 11\,k\Omega$$

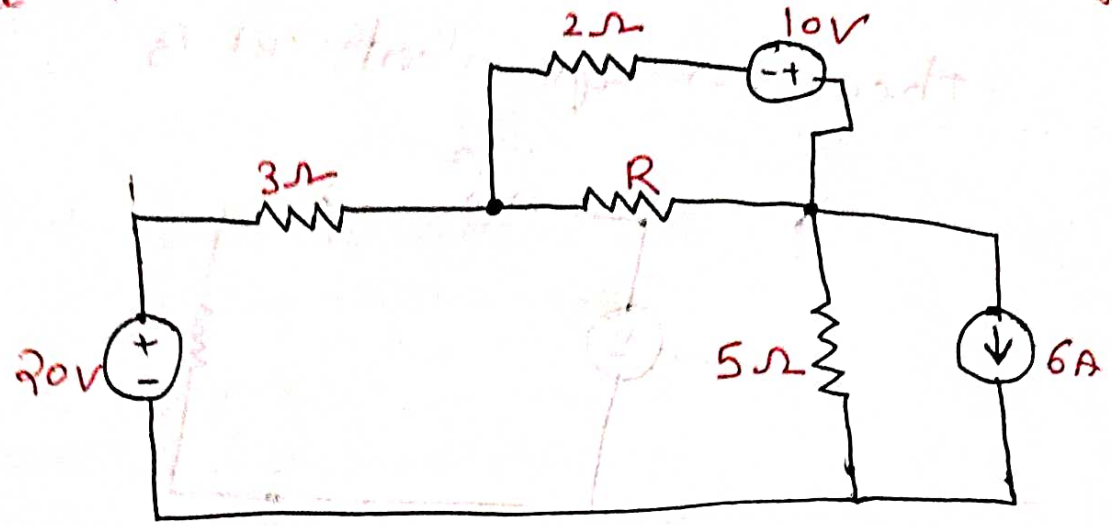
$$\therefore V_{R_L} = 19.5\left[\frac{11\,k\Omega}{(11\,k\Omega)(11\,k\Omega)}\right]$$

$$= 9.75\,V$$

$$\therefore P_{max} = \frac{V_{Th}^2}{(R_L + R_{Th})^2} \cdot R_L = 8.64\,mW$$

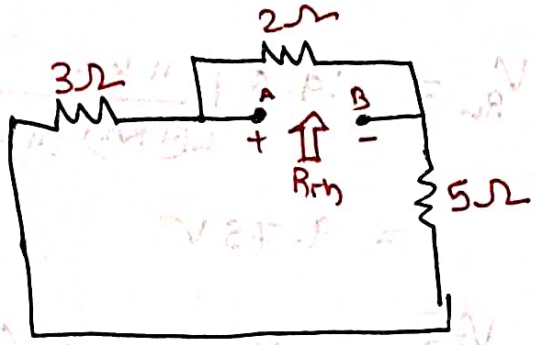
Example : 03

Find the maximum power that can be delivered to the resistor R in the circuit below.



Sol:-

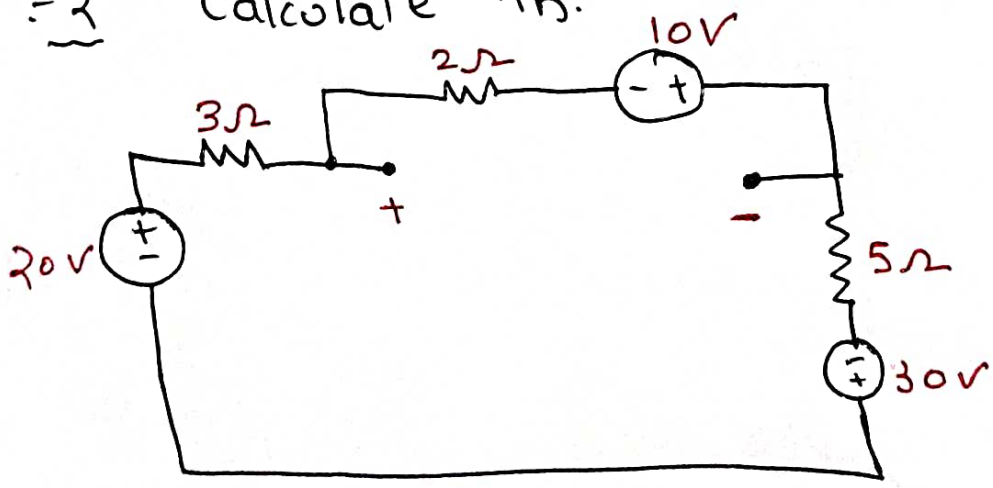
Step 1: Calculate R_{Th}:



$$R_{Th} = (3+5) \| 2$$

$$= 8 \| 2 = 1.6\Omega$$

Step :2 Calculate V_{Th}:



$$\Rightarrow 20 - 3i_1 - 2i_1 + 10 + 30 - 5i_1 = 0$$

$\Rightarrow \quad 60 = 10 i_1$

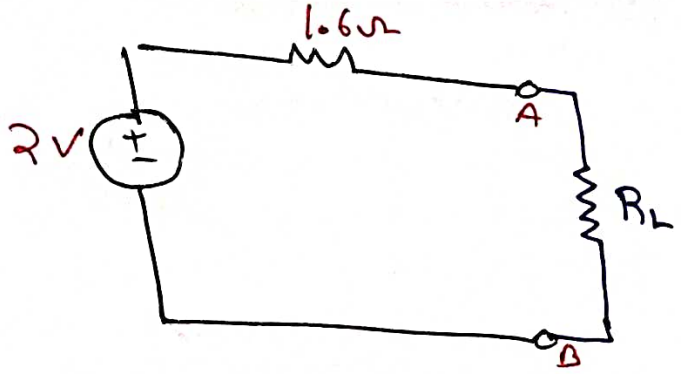
$\Rightarrow \quad \boxed{i_1 = 6A}$

*. $\quad 10 + V_{Th} - 2 i_1 = 0$

$\Rightarrow \quad V_{Th} = 12 - 10$

$\Rightarrow \quad \boxed{V_{Th} = 2V}$

$\therefore$ The Thevenin equivalent circuit is



For maximum power $R_L = 1.6\Omega$

$$\therefore P = \frac{V_{Th}^2}{(R_L + R_{Th})^2} \cdot R_L = 625\ mW$$